

Wei Chen

chenwei8994@126.com | github.com/CWhhhaaha
+86 19980818994 | International: +1 6085093756

Education

University of Wisconsin–Madison

Aug 2024 – Dec 2025

M.S. in Electrical and Computer Engineering

Machine Learning and Signal Processing (MLSP), GPA: 3.73 / 4.00

Sichuan University

Sep 2020 – Jun 2024

B.S. in Flight Vehicle Control and Information Engineering

GPA: 3.79 / 4.00

Selected Projects

My Transformer

[GitHub](#)

- Built a from-scratch PyTorch encoder-decoder Transformer for machine translation, including multi-head attention, positional encoding, training, inference, and evaluation.
- Implemented data preparation, greedy / beam-search decoding, BLEU evaluation, checkpointing, testing, and attention visualization.

Global-Market-Newsboard

[GitHub](#)

- Developed an open-source market news dashboard focused on turning real-time information streams into a more usable interface for retail traders.
- Built a user-facing workflow spanning news aggregation, data handling, presentation logic, and practical decision-support design; deployed at globalnewsboard.cn (demo password: 123456).

Campus Lane Detection

[GitHub](#)

- Built a course project around ultra-fast deep lane detection in PyTorch using a structure-aware lane detection pipeline with training, evaluation, visualization, and speed-testing scripts.
- Adapted TuSimple / CULane-style tooling to campus driving footage and produced a 1080p / 30fps curved-lane overlay demo.

Research Manuscripts

- *Basis-Mediated Bilinear Attention: A New Method for Greatly Reducing Query–Key Pathway Parameters*. Submitted to NeurIPS 2026. Proposes BMB/BMB-UV for query-key reparameterization in 7B-scale pretraining, reducing QK parameters by up to 95.83% while preserving model quality.
- *Revisiting Positional Priors in Vision Transformers with Semantic Gating*. Manuscript in preparation on semantic gating beyond static geometry-driven bias in ViT attention.
- *Beyond Positional Encoding: Layerwise Attention Reconstruction via Query-Key Structural Priors*. Manuscript in preparation on layerwise attention reconstruction using query-key structural priors.

Skills

- **Distributed Pretraining:** PyTorch DDP, FSDP, ZeRO, DeepSpeed, Megatron-LM, mixed precision, gradient checkpointing
- **Memory-Efficient Attention:** FlashAttention, SDPA, MQA, GQA, MLA, KV-cache compression, quantization, activation checkpointing
- **Optimizers and Scaling:** AdamW, Lion, Shampoo, SOAP, Muon, QK-Clip; Chinchilla scaling laws, DataComp-LM, data mixture design
- **Post-Training:** SFT, RLHF, DPO-style preference optimization, LoRA, DoRA, QLoRA, instruction tuning, synthetic data refinement
- **Agents and Tool Use:** ReAct, Toolformer, DSPy, GEPA, Codex, Claude Code, Kimi, Swarm, multi-agent workflows

Selected Coursework

- Large Language Models and Deep Learning: [Advanced Topics in Large Language Models \(ECE 901\)](#); [Introduction to Artificial Neural Networks \(CS 539\)](#)
- Machine Learning and Optimization: [Machine Learning \(ECE 760\)](#); [Matrix Methods in Machine Learning \(ECE 532\)](#); [Topics in Mathematical Data Science \(ECE 888\)](#)
- Perception and Signal Processing: [Image Processing \(ECE 533\)](#)